Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; glm-4.7

Abstract

This report evaluates `z-ai/glm-4.7` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Observed cost in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie; OpenRouter leads Throughput in all routing modes; Infron leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes.

Overall, Infron's cross-mode strengths are E2E latency, observed cost, while OpenRouter's cross-mode strengths are throughput, Streaming TTFT, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform-level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput	Observed cost	E2E latency	Streaming TTFT	Cache hit rate
OpenRouter wins 4/4 Max advantage 17.41%, higher is better	Infron wins 3/4 Max advantage 3.97%, lower is better	Infron wins 4/4 Max advantage 8.39%, lower is better	OpenRouter wins 4/4 Max advantage 89.31%, lower is better	OpenRouter wins 3/4 Max advantage 2.79%, higher is better

Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 17.41%	Infron advantage 3.97%	Infron advantage 8.39%	OpenRouter advantage 89.31%	OpenRouter advantage 2.79%
Price First price	OpenRouter advantage 12.90%	Infron advantage 0.79%	Infron advantage 3.35%	OpenRouter advantage 14.37%	OpenRouter advantage 0.16%
Latency First latency	OpenRouter advantage 27.40%	OpenRouter advantage 105.88%	Infron advantage 0.03%	OpenRouter advantage 18.74%	OpenRouter advantage 2.66%
TTFT First ttft	OpenRouter advantage 12.11%	Infron advantage 3.22%	Infron advantage 1.51%	OpenRouter advantage 40.25%	Infron advantage 0.17%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second `usage.prompt_tokens` deltas are limited to 50 tokens within each `sort/group/round` pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in 3/4 routing modes; OpenRouter leads in 1/4; 0/4 tie	Overall metrics and provider drill-down
Performance	OpenRouter leads Throughput in all routing modes; Infron leads E2E latency in all routing modes; OpenRouter leads Streaming TTFT in all routing modes	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**Infron**	**OpenRouter**	**Infron**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**Infron**	**OpenRouter**	**Infron**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)
Latency First	Minimize full-response waiting time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first-token time	**Infron**	**Infron**	**OpenRouter**	**Infron**	**OpenRouter**	Infron leads overall (3/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference-platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first-token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token-level cache hit rate for repeated stable prefixes.	Second-request cache-read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response-returned `usage.prompt_tokens` as the input-token control while allowing small cross-platform accounting variance up to 50 tokens.
- Extends prompt-caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

2. Experimental Design, Dataset, and Controls

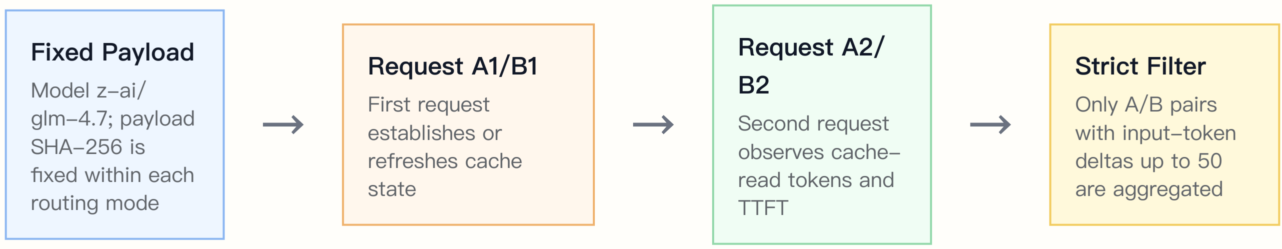
2.1 Dataset Construction

The dataset is `business_representative`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables

Figure 1: Experimental Design and Strict A/B Pairing Filter



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with `cache_read_tokens > 0`	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	`z-ai/glm-4.7`
Provider model IDs	infron: `z-ai/glm-4.7` ; openrouter: `z-ai/glm-4.7`
Platforms	Infron and OpenRouter
API protocol	`/v1/chat/completions`
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	`short` ≈1500, `medium` ≈8000, `long` ≈32000
Excluded records	16

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	196	6340852	87.70%	**$2.35911200**	2.61 tok/s	**4815.56 ms**	4237.16 ms
Throughput First	**OpenRouter**	196	6340852	**90.15%**	$11.72252143	**48.12 tok/s**	45202.33 ms	**2238.17 ms**
Price First	Infron	197	6332778	97.16%	**$1.29962000**	3.39 tok/s	**3695.85 ms**	3278.24 ms
Price First	**OpenRouter**	197	6332778	**97.32%**	$1.30994008	**47.05 tok/s**	16059.34 ms	**2866.39 ms**
Latency First	Infron	200	6414682	96.85%	$3.45964100	4.03 tok/s	**3119.38 ms**	2835.38 ms
Latency First	**OpenRouter**	200	6414682	**99.43%**	**$1.68043559**	**5.13 tok/s**	3120.18 ms	**2387.82 ms**
TTFT First	**Infron**	199	6396238	**90.65%**	**$1.47298600**	3.17 tok/s	**3927.90 ms**	3460.80 ms
TTFT First	OpenRouter	199	6396238	90.50%	$6.21110423	**41.60 tok/s**	9840.11 ms	**2467.55 ms**

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	3205.84 ms	**15260.63 ms**	**27522.80 ms**	2755.11 ms	14748.86 ms	26879.53 ms
Throughput First	**OpenRouter**	**1986.21 ms**	488985.92 ms	978823.46 ms	**1844.43 ms**	**5059.73 ms**	**5890.09 ms**
Price First	Infron	**3175.76 ms**	**8392.18 ms**	**12749.50 ms**	2778.86 ms	7766.03 ms	12389.07 ms
Price First	**OpenRouter**	3223.35 ms	22306.38 ms	597046.76 ms	**2420.72 ms**	**5461.09 ms**	**8349.43 ms**
Latency First	Infron	**2356.66 ms**	6139.54 ms	10955.27 ms	**2168.31 ms**	5550.92 ms	10530.32 ms
Latency First	**OpenRouter**	2896.61 ms	**5714.99 ms**	**7332.56 ms**	2183.62 ms	**4346.92 ms**	**5286.50 ms**
TTFT First	**Infron**	2817.32 ms	10246.12 ms	**29526.18 ms**	2313.33 ms	9510.77 ms	29429.36 ms
TTFT First	OpenRouter	**2542.11 ms**	**6319.66 ms**	55239.59 ms	**2048.63 ms**	**5144.78 ms**	**10020.13 ms**

Mean deltas use bootstrap 95% CIs; p–values use paired sign–flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p–value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	**80773.54 ms**	48187.49 ms to 115069.98 ms	<0.001	196	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	**–3997.97 ms**	–5040.73 ms to –3118.27 ms	<0.001	196	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	**–13.0327 tok/s**	–15.4306 tok/s to –10.7098 tok/s	<0.001	196	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	**$0.04777250**	$0.04158206 to $0.05474179	<0.001	196	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	**–3.91 pp**	–8.94 pp to 1.55 pp	0.1547	196	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	**24726.98 ms**	9517.45 ms to 42808.20 ms	<0.001	197	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	**–823.70 ms**	–1358.63 ms to –275.83 ms	0.0030	197	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	**–5.0436 tok/s**	–7.0884 tok/s to –3.2296 tok/s	<0.001	197	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	**$0.00005239**	$–0.00230687 to $0.00269302	0.9660	197	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	**–3.25 pp**	–5.89 pp to –0.73 pp	0.0140	197	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	**1.61 ms**	–607.85 ms to 509.31 ms	0.9955	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	**–895.11 ms**	–1460.28 ms to –439.17 ms	<0.001	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p–value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	**–0.5847 tok/s**	–1.0223 tok/s to –0.1315 tok/s	0.0177	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	**$–0.00889603**	$–0.01153673 to $–0.00616604	<0.001	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	**–1.46 pp**	–3.96 pp to 1.22 pp	0.3007	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	**11824.42 ms**	674.68 ms to 27150.73 ms	0.1325	199	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	**–1986.49 ms**	–2989.55 ms to –1004.35 ms	<0.001	199	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	**–3.1953 tok/s**	–4.5543 tok/s to –2.1213 tok/s	<0.001	199	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	**$0.02380964**	$0.01799208 to $0.02969512	<0.001	199	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	**1.13 pp**	–3.76 pp to 6.03 pp	0.6888	199	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	**0**	**0.0000**	**0**	**0.00 ms**	4237.16 ms	**4815.56 ms**
Throughput First	**OpenRouter**	852412	2174.5204	392	2238.17 ms	**2238.17 ms**	45202.33 ms
Price First	Infron	**0**	**0.0000**	**0**	**0.00 ms**	3278.24 ms	**3695.85 ms**
Price First	**OpenRouter**	300492	762.6701	394	2866.39 ms	**2866.39 ms**	16059.34 ms
Latency First	Infron	**0**	**0.0000**	**0**	**0.00 ms**	2835.38 ms	**3119.38 ms**
Latency First	**OpenRouter**	9454	23.6350	400	2387.82 ms	**2387.82 ms**	3120.18 ms
TTFT First	**Infron**	**0**	**0.0000**	**0**	**0.00 ms**	3460.80 ms	**3927.90 ms**
TTFT First	OpenRouter	164853	414.2035	398	2467.55 ms	**2467.55 ms**	9840.11 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions`; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
`chat_completions`	`/v1/chat/completions`	Infron	800	1600	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**
`chat_completions`	`/v1/chat/completions`	OpenRouter	800	1600	99.94%	99.69%	99.69%	99.69%	**100.00%**

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

Figure 3–E2: Normalized capability contour

Five metrics are normalized to 0–100, where higher is better.

Figure 3–E3: Cost-cache efficiency plane

X-axis is token cache hit rate; Y-axis is observed total cost.

Figure 3–E1: Routing-mode core metric comparison

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache-domain and performance differences.

Routing-Mode Drill-Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E3: Latency First core metric comparison

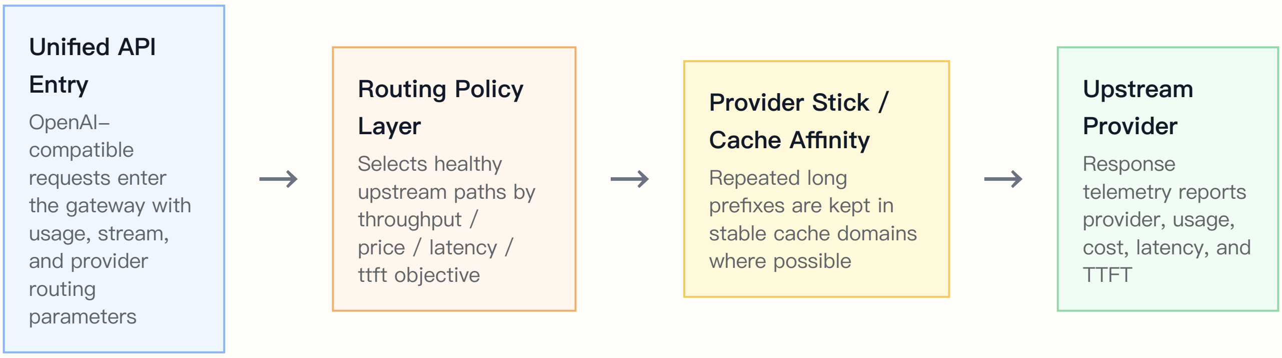
Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism

Figure 12: Infron Multi-Provider Routing and Cache Control Plane



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	392	392	`atlas-cloud` 128, `alibaba/cn` 96, `z-ai` 85, `deepinfra` 83

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	392	392	`Cerebras` 275, `StreamLake` 43, `DeepInfra` 40, `Google` 34
Price First	Infron	394	394	`alibaba/cn` 262, `atlas-cloud` 114, `deepinfra` 18
Price First	OpenRouter	394	394	`DeepInfra` 373, `StreamLake` 21
Latency First	Infron	400	400	`cerebras` 224, `atlas-cloud` 176
Latency First	OpenRouter	400	400	`DeepInfra` 336, `Cerebras` 61, `Google` 3
TTFT First	Infron	398	398	`cerebras` 190, `deepinfra` 183, `byteplus` 25
TTFT First	OpenRouter	398	398	`Cerebras` 184, `DeepInfra` 149, `Google` 57, `StreamLake` 5, `Venice` 3

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/ second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	`atlas-cloud`	128	32.65%	57/71	71	3015.57 ms	3378.43 ms	2172789	1676
Throughput First	Infron	`alibaba/cn`	96	24.49%	55/41	55	3523.03 ms	3881.86 ms	1343164	1184
Throughput First	Infron	`z-ai`	85	21.68%	45/40	54	5791.82 ms	6227.12 ms	1386006	1025
Throughput First	Infron	`deepinfra`	83	21.17%	39/44	53	5354.90 ms	6666.21 ms	1438893	1049
Throughput First	OpenRouter	`Cerebras`	275	70.15%	137/138	163	1784.62 ms	1848.12 ms	4030603	4400
Throughput First	OpenRouter	`StreamLake`	43	10.97%	23/20	42	4411.22 ms	394378.19 ms	1064769	847035
Throughput First	OpenRouter	`DeepInfra`	40	10.20%	18/22	22	1993.00 ms	2727.93 ms	213119	640
Throughput First	OpenRouter	`Google`	34	8.67%	18/16	31	3446.75 ms	4226.42 ms	1032361	544
Price First	Infron	`alibaba/cn`	262	66.50%	132/130	139	3374.68 ms	3753.99 ms	4201868	3192
Price First	Infron	`atlas-cloud`	114	28.93%	56/58	65	2982.92 ms	3405.47 ms	1840318	1534
Price First	Infron	`deepinfra`	18	4.57%	9/9	9	3744.93 ms	4688.60 ms	290592	204

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Price First	OpenRouter	`DeepInfra`	373	94.67%	188/185	196	2684.14 ms	3596.74 ms	5907457	5968
Price First	OpenRouter	`StreamLake`	21	5.33%	9/12	20	6103.59 ms	237418.80 ms	425321	291735
Latency First	Infron	`cerebras`	224	56.00%	115/109	129	1669.54 ms	1820.29 ms	1423410	2784
Latency First	Infron	`atlas-cloud`	176	44.00%	85/91	105	4319.17 ms	4772.76 ms	4991272	2238
Latency First	OpenRouter	`DeepInfra`	336	84.00%	166/170	174	2524.43 ms	3381.79 ms	5897735	5376
Latency First	OpenRouter	`Cerebras`	61	15.25%	33/28	38	1655.31 ms	1714.24 ms	496719	976
Latency First	OpenRouter	`Google`	3	0.75%	1/2	3	1982.30 ms	2407.21 ms	20228	48
TTFT First	Infron	`cerebras`	190	47.74%	98/92	113	1579.81 ms	1718.75 ms	1098069	2250
TTFT First	Infron	`deepinfra`	183	45.98%	86/97	109	3299.62 ms	4156.87 ms	4602829	2342
TTFT First	Infron	`byteplus`	25	6.28%	15/10	22	18936.10 ms	19041.43 ms	695340	367
TTFT First	OpenRouter	`Cerebras`	184	46.23%	93/91	113	1996.63 ms	2064.39 ms	2088233	2944
TTFT First	OpenRouter	`DeepInfra`	149	37.44%	74/75	87	2665.34 ms	3745.02 ms	2639463	2384
TTFT First	OpenRouter	`Google`	57	14.32%	28/29	47	2900.58 ms	3388.18 ms	1481877	912
TTFT First	OpenRouter	`StreamLake`	5	1.26%	3/2	5	7491.29 ms	553229.52 ms	130771	156626
TTFT First	OpenRouter	`Venice`	3	0.75%	1/2	3	4926.22 ms	6411.97 ms	55894	48

7.1 Cache-Rate and Cost Divergence Drill-Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing-mode differences.

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	-2.45 pp	**0.20x**	**`atlas-cloud` 32.65%**	**`Cerebras` 70.15%**	**-852412**	Cache and cost move in different directions; evaluate with speed metrics
Price First	-0.16 pp	**0.99x**	**`alibaba/cn` 66.50%**	**`DeepInfra` 94.67%**	**-300492**	Cache and cost move in different directions; evaluate with speed metrics
Latency First	-2.58 pp	2.06x	`cerebras` 56.00%	**`DeepInfra` 84.00%**	**-9454**	OpenRouter has higher cache and lower cost; provider/cache-domain mix is the main signal
TTFT First	**+0.15 pp**	**0.24x**	**`cerebras` 47.74%**	`Cerebras` 46.23%	**-164853**	Infron leads both cache and cost

8. Stratified Results: Prompt-Length Cache Performance

This section aggregates second-request cache-read tokens, token-level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt-length tier. Bold cells mark the advantaged side within each tier.

Prompt-Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
`short`	1500	Infron	**266**	**468570**	426208	90.96%	**$0.40699000**	**2062.10 ms**	1753.08 ms
`short`	1500	OpenRouter	**266**	**468570**	**438400**	**93.56%**	$1.14530131	6198.48 ms	**1687.32 ms**
`medium`	8000	Infron	**263**	**2427115**	2302784	94.88%	**$1.80182600**	**3563.12 ms**	3122.63 ms
`medium`	8000	OpenRouter	**263**	**2427115**	**2349728**	**96.81%**	$5.54600964	15638.37 ms	**2252.96 ms**
`long`	32000	Infron	**263**	**9846590**	9133536	92.76%	**$6.38254300**	**6052.63 ms**	5492.40 ms
`long`	32000	OpenRouter	**263**	**9846590**	**9234880**	**93.79%**	$14.23269038	33626.91 ms	**3538.45 ms**

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
`short`	Throughput First	85.41%	**92.62%**	**OpenRouter**

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
`short`	Price First	91.77%	**97.42%**	OpenRouter
`short`	Latency First	93.06%	**94.03%**	OpenRouter
`short`	TTFT First	**93.62%**	90.29%	Infron
`medium`	Throughput First	91.44%	**93.48%**	OpenRouter
`medium`	Price First	94.34%	**99.82%**	OpenRouter
`medium`	Latency First	98.21%	**98.38%**	OpenRouter
`medium`	TTFT First	95.33%	**95.34%**	OpenRouter
`long`	Throughput First	86.93%	**89.25%**	OpenRouter
`long`	Price First	**98.13%**	96.67%	Infron
`long`	Latency First	96.70%	**99.96%**	OpenRouter
`long`	TTFT First	**89.35%**	89.31%	Infron

9. Stratified Results: Group-Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	48	48	84.52%	**$0.20900000**	**15394.45 ms**	14543.95 ms
Infron	2	49	49	**79.53%**	**$0.41033100**	**22335.58 ms**	21558.06 ms
Infron	3	49	49	92.16%	**$0.92022400**	**13146.75 ms**	12940.77 ms
Infron	4	50	50	94.70%	**$0.81955700**	**7774.10 ms**	7350.46 ms
OpenRouter	1	48	48	**93.31%**	$2.92280559	487720.79 ms	**5337.85 ms**
OpenRouter	2	49	49	77.90%	$2.62445875	623230.31 ms	**5314.08 ms**
OpenRouter	3	49	49	**94.84%**	$2.97944500	28482.04 ms	**4926.91 ms**
OpenRouter	4	50	50	**94.93%**	$3.19581209	44049.73 ms	**4526.62 ms**

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	48	48	91.88%	**$0.16758700**	**9304.59 ms**	8268.21 ms
Infron	2	50	50	99.12%	**$0.33531200**	**7282.33 ms**	7144.26 ms
Infron	3	49	49	**98.17%**	$0.40023700	**5469.66 ms**	5079.31 ms
Infron	4	50	50	99.08%	$0.39648400	10892.53 ms	10016.62 ms
OpenRouter	1	48	48	**94.85%**	$0.52433344	42930.43 ms	**5543.44 ms**

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	2	50	50	**99.31%**	$0.37593736	23865.89 ms	**5852.94 ms**
OpenRouter	3	49	49	95.28%	**$0.25281960**	6450.57 ms	**4602.33 ms**
OpenRouter	4	50	50	**99.68%**	**$0.15684968**	**8191.63 ms**	**5221.84 ms**

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	98.13%	$0.87834400	**5480.00 ms**	5078.45 ms
Infron	2	50	50	94.96%	$0.87516500	6819.94 ms	5605.96 ms
Infron	3	50	50	94.98%	$0.86795900	6129.32 ms	5612.93 ms
Infron	4	50	50	99.49%	$0.83817300	6073.53 ms	5363.47 ms
OpenRouter	1	50	50	**99.65%**	**$0.45228386**	5727.80 ms	**4333.76 ms**
OpenRouter	2	50	50	**98.54%**	**$0.36502139**	**5821.87 ms**	**4263.78 ms**
OpenRouter	3	50	50	**99.69%**	**$0.37872743**	**5397.15 ms**	**4378.57 ms**
OpenRouter	4	50	50	**99.89%**	**$0.48440291**	**5531.57 ms**	**4154.45 ms**

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	63.83%	**$0.80475000**	28182.96 ms	28066.61 ms
Infron	2	49	49	**98.63%**	**$0.24074900**	**6190.14 ms**	**5378.94 ms**
Infron	3	50	50	**99.75%**	**$0.20363800**	6177.89 ms	5146.51 ms
Infron	4	50	50	**99.80%**	**$0.22384900**	**5185.56 ms**	**3845.64 ms**
OpenRouter	1	50	50	**94.85%**	$1.36985113	**4181.53 ms**	**3503.71 ms**
OpenRouter	2	49	49	75.58%	$2.11139781	9203.58 ms	5565.45 ms
OpenRouter	3	50	50	99.34%	$1.05860361	**4848.89 ms**	**3330.83 ms**
OpenRouter	4	50	50	92.41%	$1.67125168	8486.81 ms	5933.69 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	Good for throughput-first tasks; constrain cost and cache separately
Price First	Minimize request and token cost	OpenRouter leads overall (3/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	Good for throughput-first tasks; constrain cost and cache separately
Latency First	Minimize full-response waiting time	OpenRouter leads overall (4/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	Good for throughput-first tasks; constrain cost and cache separately
TTFT First	Minimize streaming first-token time	Infron leads overall (3/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	Cache and cost are stronger; still check speed SLA

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**Infron**	**OpenRouter**	**Infron**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**Infron**	**OpenRouter**	**Infron**	**OpenRouter**	OpenRouter leads overall (3/5 metrics)
Latency First	Minimize full-response waiting time	**OpenRouter**	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first-token time	**Infron**	**Infron**	**OpenRouter**	**Infron**	**OpenRouter**	Infron leads overall (3/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run

Limitation	Impact	Next step	Current handling
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path

Summary	`export/glm47_all_experiments/`
	`routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions`
	`summary.json`

Paired dataset	`export/glm47_all_experiments/`
	`routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions`
	`benchmark_pairs.csv`

Request-level dataset	`export/glm47_all_experiments/`
	`routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions`
	`benchmark_requests.jsonl`

Filtered structured records	`export/glm47_all_experiments/`
	`routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions`
	`records.json`

Excluded-record audit	`export/glm47_all_experiments/`
	`routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions`
	`records_excluded.json`

Test source	`tests/test_rerun_routing_sort_cache_cost_ab.py`
Benchmark runner source	`scripts/rerun_routing_sort_cache_cost_ab.py`
HTML report renderer source	`scripts/render_glm52_deepseek_style_report.py`
Dataset reference	`business_representative` built-in representative business templates; request-level export is `benchmark_requests`